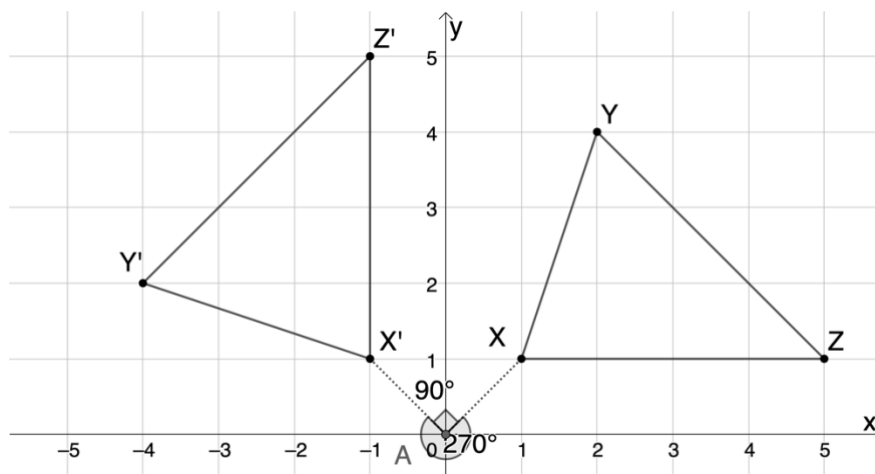


On the coordinate plane shown, $\triangle XYZ$ is rotated to create $\triangle X'Y'Z'$. Use this image for questions 1-6.



There are two ways to describe the rotation. Complete both statements using the information in the given figure.

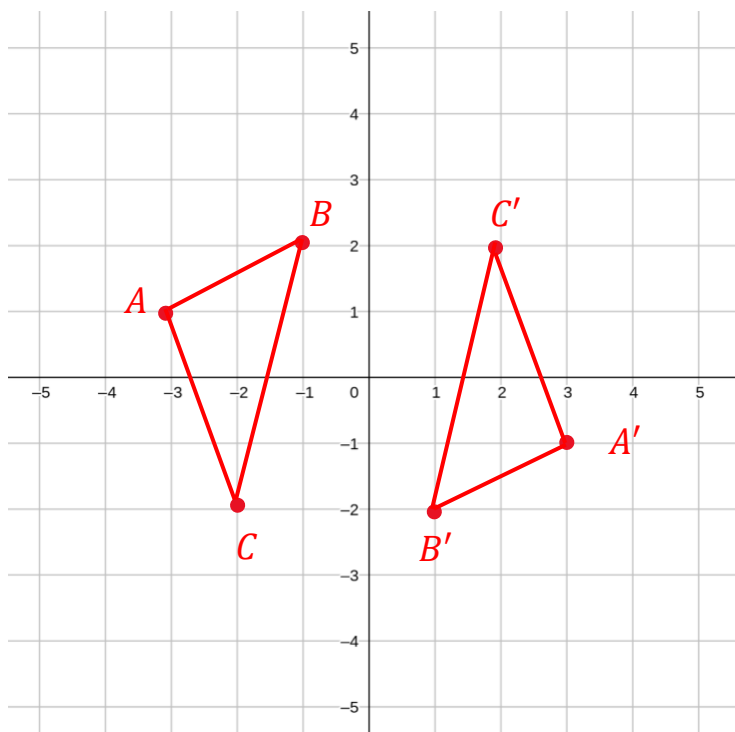
- $\triangle XYZ$ is rotated 270° clockwise about the origin to create $\triangle X'Y'Z'$.
- $\triangle XYZ$ is rotated 90° counterclockwise about the origin to create $\triangle X'Y'Z'$.

Complete the table with the coordinates of each vertex.

Coordinates		Coordinates	
3. Point X	(1,1)	Point X'	(-1,1)
4. Point Y	(2,4)	Point Y'	(-4,2)
5. Point Z	(5,1)	Point Z'	(-1,5)

- Describe the relationship between the coordinates of the preimage and the image.
The y -coordinates of the image are the same as the x -coordinates of the preimage. The x -coordinates of the image are the opposites of the y -coordinates of the preimage.

7. The vertices of $\triangle ABC$ have their coordinates given in the below table. Plot $\triangle ABC$ on the coordinate grid.



Rotate $\triangle ABC$ 180° clockwise with the origin as the center of rotation. Label the image $\triangle A'B'C'$ and complete the table for the image.

Coordinates		Coordinates	
8. Point A	$(-3, 1)$	Point A'	$(3, -1)$
9. Point B	$(-1, 2)$	Point B'	$(1, -2)$
10. Point C	$(-2, -2)$	Point C'	$(2, 2)$